

Homogena Lin NDE s konst. koef. višjega reda

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = 0$$

$$a_0, \dots, a_n \in \mathbb{R} \text{ (ali } \mathbb{C})$$

karak. en. / polinom $\rightarrow$ ničle $\lambda_1, \dots \rightarrow$ rešitev $e^{\lambda x}$

30. $y'' - y' - 2y = 0$

$$\lambda^2 - \lambda - 2 = 0$$

$$(\lambda + 1)(\lambda - 2) = 0$$

$$\rightarrow y = C_1 e^{2x} + C_2 e^{-x}$$

31. $y'' + 2y' = 0$

$$\lambda^2 + 2\lambda = 0$$

$$\lambda(\lambda + 2) = 0$$

$$\rightarrow y = C_1 + C_2 e^{-2x}$$

če ima karakt. polinom večkrat. ničlo (reda k):

rešitve $e^{\lambda x}, x e^{\lambda x}, \dots, x^{k-1} e^{\lambda x}$

32. $y'' + 6y' + 9y = 0$

$$\lambda^2 + 6\lambda + 9 = 0$$

$$(\lambda + 3)^2 = 0$$

$$y = C_1 e^{-3x} + C_2 x e^{-3x}$$

33. $y''' + 3y'' + 3y' + y = 0$

$$\lambda^3 + 3\lambda^2 + 3\lambda + 1 = 0$$

$$(\lambda + 1)^3 = 0$$

$$\rightarrow y = C_1 e^{-x} + C_2 x e^{-x} + C_3 x^2 e^{-x}$$

	1	3	3	1
-1		-1	-2	-1
	1	2	1	0

34. $y'' - 2y' - 4y = 0$

$$\lambda^2 - 2\lambda - 4 = 0$$

$$(\lambda + 2)(\lambda - 2) = 0$$

$$y = C_1 e^{-2x} + C_2 e^{2x} + C_3 x e^{2x}$$

Kompleksne ničle (polinoma z $\mathbb{R}$ koef.): $\lambda = \alpha \pm i\beta$

→ rešitvi $e^{\alpha x} \cos \beta x$ in $e^{\alpha x} \sin \beta x$

35. $y^{(4)} + 2y'' + 5y = 0$

$$\lambda^2 + 2\lambda + 5 = 0$$

$$(\lambda + 1)^2 = (2i)^2$$

$$(\lambda + 1 + 2i)(\lambda + 1 - 2i) = 0$$

$$\rightarrow \underline{y = C_1 e^x \cos(2x) + C_2 e^x \sin(2x)}$$

36. $y^{(4)} + 6y'' + 9 = 0$

$$(\lambda^2 + 3)^2 = 0$$

$$(\lambda + \sqrt{3}i)^2 (\lambda - \sqrt{3}i)^2 = 0$$

$$\rightarrow \underline{y = C_1 \cos \sqrt{3}x + C_2 \sin \sqrt{3}x + C_3 x \cos \sqrt{3}x + C_4 x \sin \sqrt{3}x}$$

Euler-Cauchyjev OČ

38. $2x^2 y'' - 7xy' + 9y = 0$

$$2\lambda(\lambda-1) - 7\lambda + 9 = 0$$

$$\lambda^2 - \frac{9}{2}\lambda + \frac{9}{2} = 0$$

$$\lambda_1 = 3$$

$$\lambda_2 = \frac{3}{2}$$

$$\rightarrow y = C_1 x^3 + C_2 x^{3/2}$$

39. $x^2 y'' - 3xy' + 4y = 0$

$$(\lambda - 2)^2 = 0$$

$$\rightarrow y = C_1 x^2 + C_2 x^2 \ln|x|$$

Variacijski račun

Ekstrem funkcionala $I(y) = \int_a^b L(x, y, y') dx$, kjer je $L: [a, b] \times \mathbb{R}^2 \rightarrow \mathbb{R}$ zvezna parc. odv. funkc., je vedno dosežen v stac. točki, tj. taki, ki zadošča Euler-Lagrangevi enačbi:

$$\underline{L_y = \frac{d}{dx} L_{y'}}$$

Včasih zahtevamo še robne pogoje $y(a) = A$ in $y(b) = B$ (enega ali oba).

Če vsaj 1 robni pogoj ni predpisan, namesto tega upoštevamo transverzalnostni pogoj:

$$L_{y'}(a) = L_{y'}(b) = 0.$$

(45) Poišči minimum funk. $I(y) = \int_1^2 (x^2 (y')^2 + 2y^2) dx$ po vseh zv. odv. funkcijah na $[1, 2]$, $y(1) = 1$ in $y(2) = 1$. Pokaži tudi, da je I na množici funkcij nesmejen navzgor.

$$L = x^2 y'^2 + 2y^2$$

$$L_y = L_{yy} \rightarrow \in L: L_y = L_{xy'} + 2x^2 y''$$

$$L_{y'} = 2x^2 y'$$

$\vdots$

$$y = Cx^2 + Dx \rightarrow \text{robni pogoji} \rightarrow C = 1, D = 0$$